

Brand Category Discovery and AI Validation

Sample results and method

Yunxiang Mo · October 2026

Summary

I built a Category → Brand table for **60 consumer categories (957 brand rows)**, starting from the Kantar BrandZ 2025 Global and China Top 100 brands, and validated every category name with the fixed question “*What are the best brands for {category}? List 10 brands.*”

- **All 60 categories pass** (Brand Recall@10 ≥ 0.8) with DeepSeek V4 Pro, after expanding the brand tables and renaming 5 categories.
- Because some brands were added after seeing DeepSeek’s answers, I froze the table and ran two **independent checks**:
 - DeepSeek re-sampled at temperature 1.0: **44/60** pass.
 - A second model, Qwen 3.7 Plus: **32/60** pass (12 more categories score 0.7, one brand short).
 - **28 categories pass both checks.**
- Total API cost for all reported runs: **about US\$0.34.**

Method

Task 1: brands → categories. Seed pool = Kantar BrandZ 2025 Most Valuable Global Brands (Top 100) + Most Valuable Chinese Brands (Top 100), 189 unique brands (88 World, 101 China). Each brand was assigned one primary category; synonymous labels were merged (30 merges), giving 66 categories. Six B2B categories (e.g. semiconductors, enterprise software) are kept but not validated, since the target is consumer-facing queries. **Unclassified seed brands: 0.**

Task 2: expand each category. For each of the 60 consumer categories I collected other major brands from public ranking pages (Wikipedia lists, Brand Finance, industry statistics). Every brand row carries a source URL. **Every category has ≥ 10 brands** (min 10, median 16, max 29). World/China = headquarters location (mainland China, Hong Kong, Macau = China).

Task 3: AI validation. Each category is asked once, in a fresh single-turn request, through OpenRouter. The answer is parsed into a brand list, and

Brand Recall@10 = (number of the AI’s first 10 brands found in the category’s table) / 10, Pass if ≥ 0.8 .

Brand names are matched after normalization (case, accents, punctuation, “Inc./Group/.com” suffixes) and an alias list for spelling variants (e.g. *Shoo Loong Kan* = *Xiaolongkan*). If the AI returns fewer than 10 brands, the missing slots count as misses.

When a category failed, I looked at the brands the AI returned that were not in the table and classified each failure:

1. **Missing brand.** The AI named a real brand of that category that my table lacked. It was added only if an independent source (mostly the Wikipedia summary) explicitly describes it as that kind of brand: 119 of 135 candidates accepted. These rows are flagged, and recall is always also reported *without* them (“pre-additions”).
2. **Misunderstood name.** The AI answered a different category. The category was renamed and re-tested (5 cases, below).

Results

Run	Model / temperature	Brand table	Pass	Mean Recall@10
Round 1, first scoring	DeepSeek V4 Pro / 0	initial	23 / 60	0.63
Round 1	DeepSeek V4 Pro / 0	initial (pre-additions)	26 / 60	0.69

Run	Model / temperature	Brand table	Pass	Mean Recall@10
Round 1	DeepSeek V4 Pro / 0	with additions	56 / 60	0.88
Round 2 (5 renamed)	DeepSeek V4 Pro / 0	with additions	5 / 5	0.96
Final (latest attempt per category)	DeepSeek V4 Pro / 0	with additions	60 / 60	—
Hold-out (table frozen)	DeepSeek V4 Pro / 1.0	with additions	44 / 60	0.80
Cross-model (table frozen)	Qwen 3.7 Plus / 0	with additions	32 / 60	0.72
Cross-model (table frozen)	Qwen 3.7 Plus / 0	pre-additions	20 / 60	0.62

Column definitions. **Run:** one pass over the categories. The first row used my first parser and alias list (later fixed: e.g. it read Markdown bullet notes as brands). It is kept for transparency, and all other rows re-score the same saved answers with the final code. **Brand table:** “pre-additions” excludes the 119 brands added after validation. **Pass:** number of categories with Recall@10 $\geq$ 0.8. **Mean Recall@10:** average over categories, range 0–1. Each category was sampled once per run; no confidence intervals were computed.

Renamed categories (round 1 → round 2, DeepSeek):

Original name	What the AI did	New name	Recall@10 after
Real Estate Developers	listed appliance/fixture brands (Sub-Zero, Miele)	Chinese Real Estate Developers	0.9
Credit Card Networks	mixed in card issuers (Chase, Capital One)	Payment Card Networks	0.9
Home Improvement Stores	listed tool/paint brands (DeWalt, Benjamin Moore)	Home Improvement Retail Chains	1.0
Cigarettes	refused to answer	Cigarette Brands	1.0
AI Chatbots	listed companies (OpenAI, Anthropic)	AI Chatbot Apps	1.0

Column definitions. **What the AI did:** my reading of the round-1 answer. **Recall@10 after:** round-2 score with additions. For AI Chatbots the rename alone did not help: the answer still listed companies, and it passed only after mapping company → product (OpenAI → ChatGPT). That mapping is my judgment.

Sample of the master table (full table: 957 rows, attached):

Category	Brand	World/China	Result
Smartphones	Apple / Samsung / Google Pixel	World	Pass
Smartphones	Xiaomi / Huawei / OPPO / vivo	China	Pass
Baijiu	Moutai / Wuliangye / Luzhou Laojiao	China	Pass
Electric Vehicles	Tesla / BMW	World	Pass
Electric Vehicles	BYD / Li Auto / NIO	China	Pass
Bubble Tea Chains	Mixue Bingcheng / Heytea / Chagee	China	Pass

Column definitions. One row per brand in the real table; several brands are shown per line here only to save space. **Result:** validation result of the category’s latest attempt.

What the checks show

- *Observed:* the brand additions also help on new answers. In the DeepSeek hold-out, passes rise from 29 to 44 with the additions; with Qwen, from 20 to 32. So they are not fitted to one single answer, but the 60/60 is optimistic.
- *Observed:* the two models agree on 40 of 60 categories: 28 pass both, 12 fail both. The categories failing both are broad food and drink (Dairy Products, Beer, Energy Drinks, Processed Meat Products, Hot Pot Restaurant Chains), services (Insurance Companies, Hotel Chains, Supermarket Chains, Online Healthcare Platforms), Cosmetics, Drones and Traditional Chinese Medicine.

- *Observed, Qwen*: “Online Shopping Platforms” was read as e-commerce *software* (Shopify, WooCommerce; recall 0.0), and “AI Chatbot Apps” partly as business chatbot builders (Intercom, ManyChat). These are naming problems a single model did not reveal.
- *Interpretation*: most other misses are real brands outside my table: US regional or niche names (Mint Mobile, USAA, GEICO), premium or craft picks (Aman, Westmalle), and DTC beauty (Rare Beauty). Broad category names seem to invite a different “long tail” each time. Narrower names, e.g. by sub-type or market, are the likely fix. This is not yet tested.

Limitations

- One sample per category per run; no variance or confidence interval was computed.
- Brand additions and some aliases were made after seeing round-1 answers. This is mitigated by reporting pre-additions recall and by the frozen-table hold-out and cross-model runs.
- About half of the Task 2 rows are supported by search-result snippets, not by full pages. World/China labels and “clearly belongs to the category” are manual judgments.
- Both validators answered with a US-centric bias, which affects the World/China balance of what counts as “best”.

Next steps toward 1,000 categories

1. **Larger seed pool**: Kantar and Brand Finance sector rankings plus e-commerce category trees.
2. **Semi-automated Task 2**: candidate brands from several sources, with human spot checks.
3. **Multi-model validation by default**: a category passes only if it passes on at least two models. The categories that fail on both models get narrower names.

The code, raw AI answers, evidence logs and per-category results are in the repository:

<https://github.com/moyunxiang/ai-observatory>. Every number here can be recomputed from the saved answers.

Appendix: per-category results

#	Category	Brands (W/C)	DeepSeek r1 pre → with	Final	Hold-out	Qwen
1	Smartphones	14 (5/9)	0.7 → 1.0	Pass	0.8	0.9
2	Search Engines	12 (9/3)	1.0 → 1.0	Pass	0.9	0.9
3	Social Media Platforms	17 (10/7)	0.9 → 0.9	Pass	1.0	0.8
4	Short Video Apps	15 (10/5)	0.7 → 0.9	Pass	0.9	1.0
5	Video Streaming Services	16 (11/5)	1.0 → 1.0	Pass	1.0	0.9
6	Music Streaming Services	14 (11/3)	0.6 → 1.0	Pass	1.0	0.9
7	Online Shopping Platforms	16 (9/7)	0.4 → 0.9	Pass	0.8	0.0
8	Fast Food Chains	18 (18/0)	0.7 → 1.0	Pass	1.0	0.9
9	Coffee Chains	17 (15/2)	0.7 → 1.0	Pass	0.9	0.7
10	Bubble Tea Chains	18 (11/7)	0.7 → 0.9	Pass	0.9	0.7
11	Hot Pot Restaurant Chains	13 (0/13)	0.7 → 1.0	Pass	0.7	0.5
12	Soft Drinks	14 (13/1)	0.7 → 1.0	Pass	0.9	0.9
13	Bottled Water	20 (17/3)	0.5 → 0.8	Pass	0.9	0.7
14	Dairy Products	15 (13/2)	0.7 → 0.9	Pass	0.2	0.3
15	Baijiu	17 (0/17)	0.9 → 0.9	Pass	0.9	0.9
16	Beer	23 (17/6)	0.1 → 1.0	Pass	0.6	0.6
17	Energy Drinks	18 (16/2)	0.5 → 0.8	Pass	0.7	0.5

#	Category	Brands (W/C)	DeepSeek r1 pre → with	Final	Hold-out	Qwen
18	Soy Sauce	17 (12/5)	0.6 → 1.0	Pass	0.2	0.9
19	Processed Meat Products	16 (14/2)	0.6 → 1.0	Pass	0.3	0.6
20	Banks	29 (19/10)	1.0 → 1.0	Pass	1.0	0.4
21	Payment Card Networks	11 (10/1)	0.4 → 0.4 (renamed: 0.9 → 0.9)	Pass	0.9	0.9
22	Digital Payment Services	12 (10/2)	0.9 → 0.9	Pass	0.9	0.6
23	Insurance Companies	21 (15/6)	0.8 → 0.8	Pass	0.7	0.4
24	Health Insurance	11 (11/0)	0.8 → 0.8	Pass	0.8	0.8
25	Mobile Carriers	21 (18/3)	1.0 → 1.0	Pass	0.9	0.3
26	Internet Service Providers	12 (12/0)	0.8 → 0.8	Pass	0.9	1.0
27	Airlines	24 (19/5)	0.9 → 0.9	Pass	1.0	0.9
28	Express Delivery Services	17 (10/7)	0.8 → 1.0	Pass	0.7	0.9
29	Food Delivery Platforms	14 (12/2)	0.8 → 0.8	Pass	0.9	0.8
30	Ride-Hailing Apps	10 (9/1)	0.9 → 0.9	Pass	0.9	0.8
31	Online Travel Agencies	17 (13/4)	0.7 → 1.0	Pass	0.9	0.7
32	Hotel Chains	22 (19/3)	0.7 → 1.0	Pass	0.5	0.4
33	Restaurant Review Platforms	12 (11/1)	0.9 → 0.9	Pass	1.0	0.8
34	Real Estate Agencies	15 (13/2)	0.8 → 1.0	Pass	0.9	0.8
35	Chinese Real Estate Developers	12 (0/12)	0.0 → 0.0 (renamed: 0.8 → 0.9)	Pass	0.8	0.7
36	Electric Vehicles	19 (11/8)	1.0 → 1.0	Pass	0.9	0.9
37	Cars	16 (14/2)	0.9 → 0.9	Pass	0.8	0.7
38	Luxury Fashion	17 (17/0)	0.7 → 1.0	Pass	1.0	0.7
39	Sportswear	14 (12/2)	0.6 → 1.0	Pass	0.8	0.8
40	Fast Fashion	13 (12/1)	0.9 → 0.9	Pass	1.0	0.8
41	Cosmetics	18 (17/1)	0.9 → 0.9	Pass	0.5	0.6
42	Jewelry	19 (12/7)	0.4 → 1.0	Pass	1.0	0.8
43	Home Appliances	19 (15/4)	0.6 → 1.0	Pass	0.8	0.7
44	Air Conditioners	13 (10/3)	0.7 → 0.9	Pass	0.7	0.8
45	TVs	10 (7/3)	0.8 → 0.8	Pass	0.9	0.8
46	Laptops	11 (9/2)	0.8 → 0.8	Pass	0.9	0.9
47	Drones	12 (3/9)	0.8 → 0.8	Pass	0.5	0.7
48	Video Game Consoles	16 (14/2)	0.6 → 1.0	Pass	1.0	0.8
49	Consumer Electronics	15 (13/2)	0.6 → 1.0	Pass	0.8	0.7
50	AI Chatbot Apps	15 (11/4)	0.8 → 0.9 (renamed: 0.8 → 1.0)	Pass	0.9	0.5
51	Supermarket Chains	16 (15/1)	0.7 → 0.9	Pass	0.7	0.7
52	Home Improvement Retail Chains	15 (15/0)	0.5 → 0.5 (renamed: 0.8 → 1.0)	Pass	1.0	0.7
53	Furniture Stores	13 (13/0)	0.5 → 0.8	Pass	1.0	0.8
54	Traditional Chinese Medicine	12 (1/11)	0.6 → 0.8	Pass	0.6	0.2
55	Online Healthcare Platforms	18 (13/5)	0.3 → 0.8	Pass	0.6	0.6
56	Job Search Platforms	15 (9/6)	0.8 → 0.8	Pass	0.8	0.9
57	News Apps	17 (14/3)	0.7 → 1.0	Pass	0.9	0.6

#	Category	Brands (W/C)	DeepSeek r1 pre → with	Final	Hold-out	Qwen
58	Video Game Companies	24 (21/3)	0.5 → 1.0	Pass	0.8	1.0
59	Movie Studios	16 (16/0)	0.8 → 0.8	Pass	0.8	0.9
60	Cigarette Brands	14 (14/0)	0.0 → 0.0 (renamed: 0.6 → 1.0)	Pass	0.0	0.9

Column definitions. **Brands (W/C)**: rows in the brand table (World / China). **DeepSeek r1 pre → with**: round-1 Recall@10 before and after brand additions; *renamed* shows round 2 for the five renamed categories. **Final**: result of the latest attempt, which decides inclusion in the master table. **Hold-out**: DeepSeek at temperature 1.0 with the frozen table. **Qwen**: Qwen 3.7 Plus with the frozen table. All recall values are in the range 0–1, from a single sample.